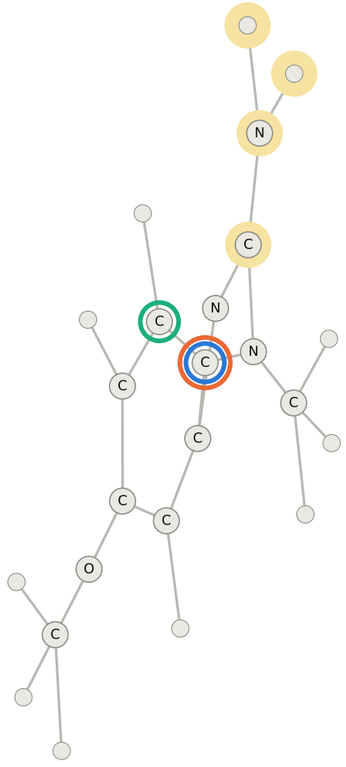
One prediction unfolded (rule model, exact). A box is a unit at a node: if 1, the fewest true literals that make it fire (n_X = count over the node and its neighbours; L0 counts atoms); if 0, its true literals fall short of the threshold. Arrows: a neighbour where the counted unit holds. Shaded box: the node differs from the prototype on that bit.

test graph 5: true mutagen · rules mutagen
rings = nodes used below (L0 aqua, L1 orange, L2 blue)



conv layer 0

L0u57 = 1 at C3
n_C = 3
n_H = 1

conv layer 1

L1u46 = 1 at C0
n_L0u57 = 1
n_NOT L0u55 = 4
n_NOT L0u11 = 4
+4 more

L1u10 = 1 at C0
n_NOT L0u11 = 4
n_NOT L0u17 = 1
n_L0u37 = 0
+3 more

conv layer 2

L2u4 = 1 at C0
n_L1u46 = 2
n_NOT L1u29 = 4
n_NOT L1u51 = 4
+4 more

L2u8 = 1 at C0
n_NOT L1u29 = 4
n_L1u34 = 0
n_NOT L1u51 = 4
+2 more

L2u11 = 0 at C0
true literals sum 0.7 < 1.1

L2u21 = 1 at C0
n_NOT L1u58 = 4
n_NOT L1u29 = 4
n_L1u46 = 2
+7 more

L2u27 = 1 at C0
n_L1u46 = 2
n_NOT L1u29 = 4
n_NOT L1u51 = 4
+4 more

L2u49 = 1 at C0
n_NOT L1u29 = 4
n_NOT L1u21 = 1
n_L1u61 = 0
+4 more

L2u50 = 1 at C0
n_NOT L1u59 = 4
n_L1u10 = 3
n_NOT L1u58 = 4
+3 more

L2u59 = 1 at C0
n_NOT L1u21 = 1
n_L1u10 = 3
n_L1u34 = 0
+6 more

prototype

P4 at C0: similarity 1.00
needs 7 units = 1, 6 = 0 (shown: 8)
¬L0u56 ∧ ¬L1u43 ∧ ¬L1u47 ∧ ¬L1u58 ∧ ~~L2u4~~ ∧ L2u8 ∧ ¬L2u11 ∧ L2u21 ∧ L2u27 ∧ ¬L2u40 ∧ L2u49 ∧ L2u50 ∧ L2u59

head

head explanation
• mean over nodes of match(P28) ≤ 0.73
• mean over nodes of match(P13) in [0.14, 0.88]
• some node is within 4 mismatches of P4
• some node is within 3 mismatches of P10

→ **"mutagen"**

PRELIMINARY